

AI and ML Specific Knowledge

1. Basic Machine Learning

- Understand core ML concepts: supervised, unsupervised, and reinforcement learning.
- Familiarity with common algorithms:
 - Regression (Linear, Logistic)
 - Decision Trees, Random Forests
 - K-Means Clustering
 - Neural Networks
- Basic implementation using scikit-learn or similar libraries.

2. Deep Learning

- Awareness of deep learning frameworks like TensorFlow, PyTorch, or Keras.
- Basic understanding of:
 - Neural Networks (CNNs, RNNs, Transformers)
 - Use cases in computer vision and natural language processing.

3. Model Serving and Deployment

- Tools for serving models:
 - **TensorFlow Serving**: For deploying models.
 - **MLflow**: For managing the ML lifecycle.
- Building RESTful APIs for model inference.
- Use of cloud-based ML services for deployment (e.g., AWS SageMaker).

4. MLOps

- **Monitoring and Logging**: Understanding tools like Prometheus, Grafana, and ELK stack for tracking ML model performance in production.
- **Model Versioning**: Using tools like DVC or MLflow.
- **Automated Retraining**: Setting up pipelines for continuous learning with new data.

Supporting Tools

1. Visualization

- Libraries: Matplotlib, Seaborn, Plotly.
- Dashboards: Tableau, Power BI, or Looker for business-level reporting.

2. Feature Stores

- Tools like Feast or Tecton for storing and serving ML features.

3. Testing in ML

- Unit and integration testing for ML models and pipelines.
 - Data quality tests and pipeline validations.
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Soft Skills and Understanding

- **Collaboration:** Work closely with data scientists, ML engineers, and business teams.
- **Domain Knowledge:** Understand the domain-specific needs of your data and ML use cases.
- **Problem Solving:** Ability to translate business challenges into scalable data solutions.